

## Study Questions: Perception

1. Close your eyes and crochet three single crochet stitches. What sensory information did you rely on? Describe the tactile and proprioceptive signals you noticed in Active Inference terms.
2. Explain how yarn tension functions as a **sensory state** at the Markov blanket. What information does it carry about the internal and external states of the crochet system?
3. An experienced crocheter notices a dropped stitch after just two stitches. A beginner does not notice until they count at the end of the row. What does this tell you about the precision of their respective generative models?
4. Describe the sensory difference between crocheting with cotton yarn and crocheting with wool yarn. How does your generative model assign different predictions to these two materials?
5. When crocheting in dim light, many crafters report relying more on touch and less on vision. Explain this shift using the concept of **precision weighting** in Active Inference.
6. You are crocheting a row of double crochets and your hook suddenly meets much more resistance than expected when pulling through a stitch. What has your perceptual system detected? What are the possible hidden causes?
7. How does the feeling of a correctly formed stitch differ from one where the hook was inserted incorrectly (e.g., into only one loop instead of both)? Describe this in terms of prediction error.
8. A crocheter describes "getting into a rhythm" during repetitive rows. How does this rhythmic perception relate to the generative model's temporal predictions?
9. When you pick up a skein of yarn in a craft store, what tactile information do you gather? How does your generative model use this information to make predictions about how the yarn will behave during crochet?

10. Explain why a brand-new yarn fiber you have never used before might increase your free energy during a project, even if the pattern is simple and familiar.
11. Some crocheters can identify the stitch type (single crochet, half double, double crochet) being used in a swatch by running their fingers over the finished fabric. What perceptual skills does this require?
12. How does the weight of the growing fabric in your lap serve as a perceptual signal? What information does it carry, and how might it update the crocheter's beliefs about progress?
13. Describe the sensory experience of **splitting the yarn** — when the hook catches only some plies of the strand instead of all of them. What prediction error does this generate?
14. A crocheter is working a complex colorwork pattern and must frequently check the pattern chart. Compare the perceptual demands of colorwork to those of a monochrome project. How does the allocation of attention shift?
15. When you are learning to maintain even tension, you must consciously attend to your finger pressure. As you gain experience, this becomes automatic. Describe this transition using the concept of perceptual inference becoming habituated.
16. A crocheter works a stitch into a stitch marker by mistake, and the marker becomes embedded in the fabric. What sensory signals might alert them to this error?
17. How does the sound of crochet (the click of the hook, the whisper of yarn sliding through fingers) contribute to perceptual monitoring? Is auditory information relevant to the crochet generative model?
18. Compare the perceptual demands of crocheting flat (back and forth in rows) versus crocheting in the round. What changes about the sensory feedback the crocheter receives?
19. When you block a finished piece (wet it and pin it to shape), you are observing how the fabric responds to water and stretching. Describe this as a perceptual inference process: what are you observing, and what beliefs are you updating?

20. Reflect on your own crochet perception: which sensory channel (vision, touch, proprioception, rhythm) do you rely on most? Has this changed as you have gained experience? How does Active Inference explain this shift?